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The impact of advanced technology and social media platforms on lifestyle medicine: a narrative review

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Abstract

Background/objectives The digital revolution-including smartphones, wearable devices, mobile applications, and online platforms- has fundamentally transformed modern human lifestyle and health behaviors. This narrative review synthesizes evidence on the positive and negative impacts of social media and smart technologies on the six pillars of lifestyle medicine: nutrition, physical activity, stress management, substance use, restorative sleep, and social connection.

Methods We conducted a narrative review of English-language, peer-reviewed literature from 2010 to 2025. Searches were performed in PubMed/MEDLINE, Scopus, Web of Science, and PsycINFO, supplemented by Google Scholar and reference-list screening.

Results Across pillars, (SMAT) can enable self-monitoring, goal-setting, personalized feedback, and remote coaching. Evidence is strongest for physical activity and dietary self-management, particularly when interventions include behaviour change techniques (e.g., prompts, gamification, and social support) and clinician input. Digital mindfulness and cognitive-behavioral programs show modest reductions in perceived stress and improvements in sleep outcomes in selected populations. However, excessive screen time and evening device use are consistently associated with poorer sleep, digital fatigue, and reduced wellbeing. Social platforms can enhance peer support and accountability, but can also amplify misinformation, unhealthy comparison, and exposure to marketing or norms that promote risky substance use.

Implications Healthcare professionals, educators, and policymakers should promote evidence-based and balanced SMAT use by assessing digital health literacy, guiding purposeful engagement, monitoring for harms such as sleep disruption and anxiety, and addressing inequities in access. Hybrid models that combine digital tools with counseling, education, and family- or community-based support appear most promising for sustained behavior change. Regulatory oversight, app validation, data privacy protections, and long-term research in diverse populations are essential to ensure safe, effective, and equitable implementation.

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1 Introduction

Advances in smart technology platforms have changed the way people interact, gather information, and monitor their health. This digital transformation offers innovative tools for health promotion but also raises new challenges across all six pillars of lifestyle medicine. Understanding the impact of social media and smart technologies on lifestyle factors is essential for clinicians, educators, and decision makers striving to optimize public health in the digital era.

When used responsibly, social media and smart technologies can be empowering. They may enable individuals to take greater control of their health, build supportive networks, and monitor key health parameters. At the same time, ethical and practical concerns arise around digital addiction, data privacy, misinformation, and the quality of online health-related content.

Wearable health devices have evolved from basic step counters to sophisticated smartwatches capable of capturing a wide range of physiological and behavioral parameters. The ability to collect real-time information about daily movement, sleep, heart rate, and stress levels has encouraged more proactive health management [1].

Chronic diseases can be prevented and managed through lifestyle interventions, which lie at the core of lifestyle medicine and are increasingly supported by smart technologies and wearables [2]. These innovations have transformed how individuals track and manage their health, promoting healthier lifestyle choices and self-management of chronic conditions [3]. Gamification elements such as step goals, activity "rings," and achievement badges may increase motivation and adherence to lifestyle change, thereby reinforcing the foundations of lifestyle medicine [4].

Although the integration of smart technologies in health tracking and monitoring, in addition to wearable health devices is a promising addition to the lifestyle medicine landscape, their potential is limited if users cannot sustain their lifestyle changes. Such tools should be add-ons, not substitutes for basic health interventions. With ongoing advancements in technology, its contribution to the future of population health and chronic disease will expand, incorporating more personalized, accessible, and effective lifestyle medicine.

However, the integration of SMAT into lifestyle medicine is not without significant concerns. Critical issues include digital addiction and compulsive use patterns, data privacy and security vulnerabilities, proliferation of health misinformation, variable quality and lack of regulation of health apps, exacerbation of health inequities through the digital divide, and potential for technology to replace rather than complement human connection and professional care.

In this narrative review, we explore how social media and advanced technologies (SMAT) influence each of the six pillars of lifestyle medicine. For each pillar, we summarize the positive and negative impacts, highlight key limitations in the evidence base, and discuss clinical and ethical implications.

2 Methodology

A targeted literature search was conducted to inform this narrative review. Searches covered January 2010 through August 2025 and included PubMed/MEDLINE, Scopus, Web of Science, and PsycINFO, supplemented by Google Scholar and manual screening of reference lists from key reviews and guidelines.

2.1 Search strategies used combinations of terms related to

- Technology Terms: "social media" OR "digital technology" OR "wearable devices" OR "mobile health" OR "mHealth" OR "health apps" OR "fitness trackers".
- Pillar-Specific Terms: "nutrition" OR "diet" OR "physical activity" OR "exercise" OR "stress management" OR "mental health" OR "substance use" OR "smoking" OR "sleep" OR "circadian" OR "social connection" OR "social support".

2.2 Inclusion criteria

English-language, peer-reviewed articles.
Systematic reviews, meta-analyses, randomized controlled trials, or large observational studies.
Studies examining SMAT in relation to at least one lifestyle medicine pillar.

2.3 Exclusion criteria

Opinion pieces or commentaries without empirical data.
Case reports.
Studies focused solely on device engineering, non-health outcomes, non-human participants, or non-health-related technology.

2.4 Screening process

Titles and abstracts of several hundred records were screened following de-duplication, with full-text review performed for articles meeting eligibility criteria. Approximately 120 articles were included in the final narrative synthesis.

Given the heterogeneity of study designs, interventions, and outcomes, a narrative synthesis approach was employed rather than quantitative meta-analysis. Findings were organized thematically by the six lifestyle medicine pillars, with cross-cutting themes identified through iterative analysis.

3 Results

3.1 SMAT impact across six pillars of lifestyle medicine

3.2 Nutrition

Mobile applications, social media, and wearable devices have enabled dietary behaviour change through smart technology. These tools support nutritional awareness and health behaviour change, but also pose risks regarding exposure to misleading or harmful content. This section elaborates on the positive and negative effects of social media and smart technology on a healthy diet.

Social media influencers can effectively alter dietary behaviour by establishing perceived social norms through sharing their experiences. They can popularize healthy eating or gluttonous food trends, depending on their content. Studies suggest influencers who post appealing images of meals and health routines foster trust and motivate their audience, particularly younger populations [5, 6].

Digital interventions using mobile apps and platforms can reduce maladaptive eating behaviors and increase healthy food consumption through feedback, education, and tracking. Strategies like meal planning, real-time feedback, and virtual support

communities are effective in reducing unhealthy dietary behaviors and increasing vegetable and fruit intake [7, 8].

Self-Monitoring and Mobile Technology enable users to track food consumption accurately and bring awareness to dietary patterns. Calorie counting, personalized feedback, educational videos, and goal tracking promote healthier eating, resulting in marked improvements in dietary behavior among diverse populations. Nutrition-specific apps offer tracking tools integrated with other health metrics, including physical activity and sleep quality [9, 10].

Wearable technology provides real-time, holistic feedback, ensuring better nutrition monitoring. It improves portion control, nutrient balance, and fosters mindful eating practices. Some devices connect with mobile apps to give real-time data on physical activity and caloric intake for behaviour change via continuous feedback loops [11].

Roles in Behavioral Reinforcement were also reported. Motivational prompts and reminders for healthy eating behaviors are provided via text messaging and gamification techniques. These interventions are effective at increasing fruit and vegetable consumption and reducing sugary drink consumption. Regular reminders via SMS, especially with goal setting and feedback, are associated with greater adherence to dietary goals and decreased intake of sugar-sweetened beverages [12, 13].

Gamification demonstrates immediate benefits for dietary behaviors in adolescents, making nutrition activities more fun and interactive. It employs game design elements like points, badges, and challenges to motivate users, especially adolescents [14].

Web-based tools like the Food4Me project indicate that personalized nutritional intervention, considering dietary, genetic, and lifestyle data, greatly improves compliance with healthy eating guidelines compared to generic advice. Personalized nutritional guidance achieves greater and longer-standing improvements in dietary quality than one-size-fits-all strategies [15].

Mobile apps and digital platforms provide swift access to nutritional insights and tools, allowing users to make informed dietary choices and take a more active role in nutrition management. These platforms make dietary information available to people who might otherwise be unable to receive it due to time, distance, geography, or nutrition education [16].

On the other hand, Influencers can also make harmful trends fashionable. While many promote healthy recipes and wellbeing, others may endorse extreme or fad diets, resulting in a range of detrimental dietary options for adherents to select from [5, 6].

Exposure to image-related content on social media, particularly among young adults, can lead to body dissatisfaction, restrictive dieting, binge eating, and consumption of unhealthy foods. Social media platforms like Instagram are rife with aestheticized body images, which can skew self-image and encourage unhealthy dieting habits, leading to eating disorder symptoms or other nutrition-related mental health issues [17].

Social media content encouraging unhealthy risk behaviors or unrealistic diets can lead to unhealthy dietary choices and emotional eating. Trends can perpetuate weight stigma and unhealthy eating patterns. Misinformation about viral diet challenges and unverified 'detox' plans promote restrictive and unbalanced eating patterns, negatively impacting metabolic health [18–20].

The enhanced accessibility of online meal delivery applications facilitates easier procurement of ready-made and processed foods, typically linked to diminished dietary

quality and elevated intake of high-fat and high-sugar items. The convenient app-based ordering of high-calorie, low-nutritional value food products may result in diminished diet quality when frequently replacing home-cooked meals [16].

Regarding younger ages, exposure to ideal body types and marketing for unhealthy foods on social media correlates with the development of eating disorders and increased consumption of unhealthy, high-calorie foods in children ages 5–18 years. Children are susceptible to marketing techniques, such as cartoon characters or influencers advertising unhealthy snacks, leading to increased consumption of processed and high-calorie foods [21].

Overall, social media and advanced technologies (SMAT) shape individual eating habits and national nutrition trends. Positively, they offer easy access to education, community support, dietary self-monitoring, and health promotion. Nonetheless, they can also promote poor eating behaviors through deceptive food advertising, body image pressures, disordered eating behaviors, and misinformation. The negative psychological and behavioral impact of curated online content and food-related peer comparison disproportionately affects young populations and vulnerable groups. Healthcare providers and policymakers should promote evidence-based digital tools that support balanced nutrition and food literacy while countering harmful content. Public education campaigns that create digital literacy and critical thinking around nutrition are central to minimizing harm and supporting healthier lifestyles.

Most of the evidence linking SMAT to dietary behaviors is derived from cross-sectional or observational studies, which limits causal inference and may be affected by self-report bias. Randomized trials of digital nutrition interventions remain relatively small, short in duration, and heterogeneous in design, making it difficult to determine which specific features (e.g., self-monitoring, social support, gamification) are most effective in producing sustained dietary change.

3.3 Physical activity

Social media and advanced technology (SMAT) have revolutionized the physical activity engagement landscape.

Technology has a significant role in supporting physical activity and exercise behavior. Active smartphone interventions have been shown to be effective for increasing moderate-to-vigorous physical activity (MVP), step counts, and decreasing sedentary behavior [22, 23]. Such tools can provide features such as real-time tracking, personalized feedback, and automated reminders, strengthening compliance with goals, and encouraging long-term behavior change [24].

Active video games (AVGs), when added to structured weight management programs, are effective in increasing physical activity and lowering measures of obesity. This may lead to improved adherence and outcomes, particularly in youth [25].

Platforms such as Facebook, Instagram, and Twitter promote behavior change through group support, sharing fitness programs, and a public commitment to goals. These platforms provide gamified experiences and encouragement narratives that promote collective action and healthy competition [26]. Mobile apps adorned with gamification features such as badges, points, and leaderboards stimulate both intrinsic and extrinsic motivators to nudge regular physical activity. These components nurture self-efficacy

and are commonly associated with increased retention and persistent adherence to exercises [27].

The Motivational Technology Model (MTM) constructs motivation-based frameworks to guide and evaluate technology-based interventions in terms of intention formation, initiation, and sustenance of behavior, thus providing a strong theoretical foundation for effective technology-based programs [27].

Fitbit, Apple Watch, and other wearables and apps provide continuous tracking of fitness metrics. Users who regularly monitor aspects of their health via any of these technologies report measurable health improvements such as reductions in waist circumference, blood pressure, and increased cardiovascular fitness [22, 23]. They have also been demonstrated to support mental well-being by increasing users' confidence, independence, and sense of achievement [28, 29].

Digital peer-support systems embedded into social media and mobile apps foster camaraderie and shared accountability. Users may challenge each other, celebrate milestones, share tips, and all of this can contribute to maintain engagement and supporting exercise frequency [30, 31].

Despite the development in this space, the overuse of fitness technology can actually stifle intrinsic motivation. Adding digital feedback may cause users to rely too much on videos instead of enjoying nature for movement, and after using tech tools, people prefer disengaging from physical activity when tech tools are not available [32].

An obsession with measuring habits, like daily steps or calories burned, can lead to anxiety or compulsive behaviors. This data-driven attitude can distract from the larger mental and emotional benefits of exercise [31]. The way social media affects physical activity is also inconsistent. Although some interventions work, others do not change behavior. However, such outcomes could be conditional on service usage, content or user characteristics [33]. In some cases, users may become discouraged by unrealistic representations of fitness or progress, resulting in comparison-based dissatisfaction or drop-out [31].

In summary, the adoption of social media and Advanced technology (SMAT) in physical activity and fitness has fundamentally changed how people interact with exercise and health behavior change. The evidence supporting wearables, fitness apps, gamification strategies, and digital peer support platforms as methods to effectively increase motivation, adherence, and physical performance across various populations is very strong. These mainly drive accountability and deliver real-time feedback that enforces healthy activity patterns and promotes long-term habit formation.

However, the adverse effects of overdependence, obsession with monitoring progress, and the potential for an unhealthy focus on unrealistic representations of fitness through social media can erode self-motivation and general well-being. In domains that relate to appearance, users may be compelled to reach algorithm-driven goals or judge themselves unfavorably against others. Personalized interventions and critical digital literacy are key to guiding people to form physical activity habits that will sustain them and make it enjoyable.

While several systematic reviews and trials support the efficacy of smartphone apps, wearables, and social media–based interventions for increasing physical activity, many studies involve highly motivated participants, short follow-up periods, and self-selected

samples. Long-term data on maintenance of behaviour change, as well as comparative effectiveness across different platforms and age groups, are still limited.

Recent umbrella reviews and meta-analyses indicate that wearable activity trackers and other digital physical activity interventions can produce small-to-moderate increases in steps and physical activity, with larger and more sustained effects when paired with behaviour change techniques and/or coaching [34, 35]. Standalone digital behaviour changes interventions also show modest benefits, but attrition is common and long-term maintenance data remain limited [36].

3.4 Stress management

In the Digital era, social media and Advanced technology (SMAT) are now being used to help manage stress and improve mental well-being. Such new technological solutions, however, bring a double effect, a positive as well as a negative one.

From therapy apps to chat-based support, digital mental-health tools have transformed care provision. They help to strip away barriers by offering individuals a sense of anonymity and accessibility, particularly for those who are reluctant to pursue traditional services. If anything, the COVID-19 pandemic proved their significance, as remote therapy sessions became more popular and accepted [37, 38].

Intelligent technologies such as cognitive enhancement (Brain Training) applications and wellness platforms have proven to be incredibly beneficial. For older adults, internet use has enhanced memory, multitasking, and cognitive flexibility. These technologies support autonomy, increase social engagement, and mitigate health disparities, especially among older adults [39].

Social media has also served mental health well. Sharing inspiring messages, connecting with a supportive group, and accessing mental-health resources have normalized distress and made people feel more connected, decreasing symptoms of depression and anxiety. While interactive social media interventions for psychological distress are in the early stages of development, engaged participation with novel applications has yielded small but statistically significant decreases in psychological distress and increases in general well-being [39, 40].

Technology-facilitated behaviour modification strategies (feedback loops, personalized goals, and mobile reminders) improve self-efficacy, emotional regulation, and motivation for self-care. The findings indicate that high engagement with humorous, empathetic or inspiring posts on social media may yield benefits for mood in adolescence and adulthood, possibly counteracting negative affect and depressive symptoms [40, 41].

Despite these advantages, however, there are unintended harms associated with becoming dependent on digital mental health. Overuse of devices and social media can result in technostress, or a psychological response to tech overload, leading to anxiety, burnout, and depression. Techno addicted profiles are characterized by a maladaptive use and a worsening of the consequences on mental health [42].

An over-reliance on technology for socialization can erode face-to-face interaction opportunities, leading to feelings of loneliness and isolation. For older individuals, the digital divide exacerbates emotional distress as a result of possibly less experience with digital tools. Persistent mental juggling, screen time, and simultaneous digital

distractions impair memory, learning, and emotional resilience; ultimately leading to cognitive rigidity and disrupted emotional regulation [16, 43, 44].

Meta-analyses have demonstrated a weak yet positive relationship between social media use and these negative outcomes, with moderating factors such as sex, age, type of platform, and intensity of use mediating the strength of the relationship [45]. The style of engagement counts as well: active users of social media may find refuge and community, while passive users may engage in self-comparison and body image dissatisfaction, with consequential risks of low self-esteem and depression [46, 47].

Visual-centric forums exacerbate this issue. Frequent exposure to idealized body images is related to disordered eating, particularly among adolescent girls [48].

Social media and digital health technologies offer facile access to mental health support, such as mindfulness apps, virtual therapy, and social connections, which may reduce stress and increase psychological resilience. Users value content that delivers emotional validation, relaxation suggestions, and peer empathy, even when formal care pathways are depleted or amidst difficult times.

Unfortunately, far too many of those experiences on an ongoing basis can leave people feeling more anxious, stressed, and less well. Problems such as technostress, digital fatigue, cyberbullying, or social comparison are risks to consider, especially for adolescents and high-frequency users. The curated online personas, information overload, and validation-seeking behaviour can have a mental toll, heightening stress, rather than lessening it.

More focus in future work is required on equitable access to high-quality mental health technologies and on promoting digital literacy, boundary-setting, and healthy screen-time activities. It will require the shared work of developers, clinicians, educators, and policymakers to design digital spaces that don't compromise but instead promote stress regulation and mental well-being.

The evidence base for digital mental health interventions is growing but remains mixed; effect sizes are often modest, and engagement tends to decline over time. Many studies rely on self-reported outcomes, and few directly compare digital tools with standard face-to-face care or hybrid models, particularly in low-resource or high-risk populations.

3.5 Risky substance use

Substance use behaviors are largely shaped by social media and smart technologies in the lives of adolescents and young adults. These tools provide youth access to content that may normalize or facilitate substance use.

Social media and mobile technology in digital health campaigns have been effective in decreasing smoking rates. A campaign conducted by the University of Waterloo doubled smoking quit rates among young adults using interactive digital tools, 32% vs. 14% [49].

Facebook interventions have demonstrated efficacy in non-daily smokers and those intending to quit. These platforms provide real-time engagement, peer support, and tailored messages, leading to increased abstinence rates [50]. Campaigns such as 'This is Quitting' and 'The Real Cost' have led to greater awareness of the risks of tobacco and have been concurrently linked with greater intentions to quit or reduce tobacco [51]. Integrating educational and regulatory information within social media can also serve to buffer youth from pro-substance use communications. Solutions such as better

moderation, age filters and digital health literacy campaigns are being recommended [52].

On the other hand, exposure to substance use promoting content on social media is also a broader public health problem. In a review, 76.3% of substance-related posts glamorize tobacco, alcohol and cannabis use [53]. Social media exposure is a predictor of usage of tobacco; adolescents that is exposed to content about such are at significantly higher risk of trying e-cigs and combustion tobacco [54, 55]. A meta-analysis of 29 studies found a strong association between the use of social media and tobacco usage [56]. There's also a dose-response relationship between screen time and drug and alcohol abuse: kids who are on social media platforms for more than 2 h a day are at markedly higher risk of smoking or vaping in comparison to non-users, risk increases with the rise of time on social media [57, 58].

Peer sharing on platforms like Instagram, TikTok, and Snapchat builds subjective social norms about drug use, in turn supporting these behaviors. By way of indirect advertising through influencers, hashtags or branded content, the tobacco and alcohol industry targets youth surreptitiously. The overwhelming amount of misleading information and the absence of social platform-level moderation provide a misleading impression of the safety or social norm of substance use [59–61].

In summary, social media and Advanced technology (SMAT) play an important role in shaping substance use addictions in young populations. These outlets have given birth to new progressive public health campaigns and peer support networks, but at the same time serve as mediums for the normalization and glamorization of drug use. A whole society response, spanning public health experts, teachers, tech platforms and policymakers, is required to promote the responsible use of technology. This may involve setting policy, boosting digital literacy and delivering an evidence-based response to the challenge of enabling innovation while providing protections.

Importantly, much of the literature on social media and Advanced technology (SMAT) and substance use focus on exposure and associations rather than rigorously evaluated interventions. There is a need for more robust experimental and longitudinal designs that can disentangle the influence of platform algorithms, peer norms, and commercial marketing from broader social determinants of substance use.

3.6 Restorative sleep

Smart technology has made it easier to track sleep and gain insight on sleep quality. Wearable technologies and smartphone applications (e.g., Fitbit, Apple Watch, Aura ring, and WHOOP) provide an easy-to-access platform for users to track their sleep duration, amount of quality sleep, and sleep disturbances. Insight from this data enables individuals to re-evaluate practices that enable good sleep hygiene, like the consumption of caffeine late in the day or erratic bedtimes [62].

Manufacturers of modern digital devices have implemented features such as blue light filters or 'Night Shift' mode that lessen the exposure to melatonin-suppressing wavelengths. While some debate still exists on their effectiveness to improve sleep, early evidence indicates that appropriate use of these agents can facilitate circadian rhythm regulation [63].

The two popular mobile operating systems—iOS by Apple and Android by Google—offer features like 'Bedtime Mode' and screen-time limits to prohibit usage of the device

during the presumed bedtime hours. These tools discourage users from pre-sleep digital stimulation, positively influencing bedtime habits and melatonin secretion [64].

Healthcare professionals and sleep experts upload educational content on sleep hygiene via video-sharing platforms such as Instagram and YouTube. These include measures such as avoiding digital usage before sleep, keeping a dark sleep environment, and having sleep patterns [65].

Conversely, there are many notable negative impacts of social media and smart technology on restorative sleep. For youth, late use and increased exposure of social media and video games increase alertness and cognitive stimulation, delaying the onset of sleep and reducing the deep stages of sleep. This has a more disruptive effect on adolescents with evening chronotypes, who are predisposed to sleep late [66–68].

Children, especially under six years of age, are showing less sleep efficiency and overall time sleeping by having nighttime screen time. In somewhat older children (5–8 years), use of screens within two hours before bed disturbed the next-day alertness, although not sleep onset latency [69, 70].

For adults, smartphone use in the hour before bed is associated with longer sleep latency, fragmented sleep, and lower sleep satisfaction. The older adults used social media to ease loneliness during the COVID-19 pandemic while they experienced reduced duration of their sleep [71, 72].

In the general population, screen exposure decreases melatonin secretions and shifts circadian phases. Moreover, content served online and particularly emotionally arousing or interactive excites the brain, making falling asleep more difficult [73–75].

Many individuals develop a behavioral pattern known as FOMO (Fear of Missing Out); furthermore, addictive interface design and persistent notifications are determinants of problematic use of social media. These lend themselves to nighttime scrolling behaviour, leading to a decreased total sleep time and less satisfaction with sleep. Meta-analyses show small but significant associations of high social media use, poor sleep quality, and increased depression and anxiety levels [76–80].

The simultaneous use of social media and Advanced technology (SMAT) has both optimistic opportunities and substantial risks to restorative sleep. On one hand, new innovations in wearable tech, sleep-tracking apps, and digital health campaigns have given people the tools to monitor and improve their sleep hygiene. The main key amongst these tools are self-awareness, positive reinforcements, and personalization to cater to a multitude of populations, including people with chronic sleep disturbances.

However, widespread (and often compulsive) usage on social platforms, particularly before sleep, still challenges delayed sleep initiation, fragmented sleep, and poor sleep satisfaction. The mechanisms by which disruption of natural circadian rhythms and sleep architecture occurs in response to blue light exposure, cognitive stimulation, emotional arousal, and continuous notifications are highlighted. That is especially true in adolescents and young adults, whose usage patterns are most extreme and whose bedtime behaviors are the most responsive to disruption.

Beyond monitoring, evidence supports structured digital therapeutics for insomnia. Recent meta-analyses of fully automated digital cognitive behavioral therapy for insomnia (dCBT-I) show clinically meaningful improvements in insomnia severity and sleep efficiency, and equivalence analyses suggest eHealth CBT-I can achieve outcomes comparable to in-person delivery in selected populations [81, 82]. These tools may be

considered for patients with insomnia symptoms, ideally with clinician oversight and attention to engagement and safety.

However, commercial sleep trackers vary in validity, and the proliferation of sleep scores may fuel 'orthosomnia'—a preoccupation with imperfect tracker results that paradoxically increases arousal and worsens sleep. Clinicians should frame tracker data as supportive (not diagnostic), and prioritize validated measures and functional outcomes over nightly fluctuations.

An updated systematic review and meta-analysis indicates that higher electronic media use is associated with poorer sleep quality and shorter sleep duration across age groups, with stronger associations when use occurs in the evening or in bed [83]. Nevertheless, most studies are cross-sectional, use self-reported screen time and sleep, and have limited adjustment for confounding (e.g., mental health, socioeconomic status, chronotype). Few trials evaluate long-term clinical outcomes, and follow-up beyond the intervention period remains limited.

3.6.1 Social connectedness

Social connectedness is intimately tied to mental and emotional well-being at all ages. Social media and smart technologies shape the ways people tend to maintain relationships, access support, and build community. While these platforms can foster a sense of belonging and reduce social isolation, they also threaten to contribute to loneliness and disconnection.

Forums and social media sites enable adolescents and socially anxious individuals to express themselves, share experiences, and keep friendships going that might be difficult to maintain otherwise. This kind of virtual self-disclosure serves to reinforce collective identity and assure peer bonding, both of which are correlated with an increased sense of social support [84].

For people with rare conditions, marginalized identities, or those residing in remote areas, online communities provide essential emotional support. These platforms allow connecting with other people who share similar difficulties, hence minimizing feelings of exclusion and ultimately enhancing social integration [85].

Research also indicates that during crises or physical distancing, including during the COVID-19 pandemic, teens were using social media not only to connect with friends but also to engage in creative expression and emotional sharing. Such interactions contributed to alleviating loneliness and bringing upon some sense of stability and belonging [86].

These digital settings, which play a significant role in our lives, support behaviour design through modelling, reinforcement, and social comparison, even as concerns about data security and misinformation persist [84].

Teens who spend more than three hours a day on social media report greater feelings of loneliness. This could be because our superficial digital connections have replaced deeper offline relationships, worsening both loneliness and depression [87].

A study of U.S. young adults showed that those in the highest quartile of social media usage had more than double the odds of perceived social isolation compared to those in the lowest quartile. The risk was three times higher for those with more frequent usage, stressing the importance of the quantity and style of engagement in determining outcomes [88].

Social comparison on platforms is often about seeing curated, idealized versions of other people's lives. This may result in dissatisfaction, envy, and low self-esteem, especially in circumstances when people feel left out of social gatherings or are unable to meet the standards set by the content [88]. The existing competitive and performative culture ingrained in some of the digital platforms can lead to emotional trauma, particularly when users feel lesser or excluded from recent social updates [89].

In older adults, the effect of media is context-dependent. Some people use it to maintain family connections, but report increased loneliness and depression, especially when face-to-face communication is reduced to digital engagement [90]. Reliance on smart technologies to mediate social contact may diminish intrinsic motivation to establish and maintain offline relationships, resulting in thinner ties and lower long-term relationship satisfaction [32].

The emergence of SMAT has fundamentally transformed how individuals' structure, define, and evaluate their social connections. Such platforms can be powerful tools for enhancing supportive interactions among peers, particularly among adolescents and marginalized populations, and for reducing the sense of isolation through online communities. Yet this comes with its hazards as well; the influence of unregulated and overuse of technologies and social media can risk perceived social isolation, poorer face-to-face communication, and more loneliness, especially if digital interactions replace deeper, real-world relationships. A key challenge is to mitigate these harms while maximizing social benefits, which needs to be a much stronger focus on responsible technology use and digital literacy from here on out.

Across studies, findings on SMAT and social connectedness are heterogeneous and appear to depend on age, baseline mental health, platform features, and the quality of interactions. Benefits are more likely when use is active, purposeful, and relationship-building (e.g., direct messaging and participation in supportive groups), rather than passive consumption.

A useful clinical distinction is active, intentional engagement (messaging, posting, participation in supportive communities) versus passive scrolling. Active use is more consistently linked with perceived support, whereas passive use is associated with social comparison and worse mood. Harms such as cyberbullying, harassment, and exposure to appearance-focused content can further erode belonging, particularly for adolescents.

Recent umbrella reviews and meta-analyses suggest that associations between social media use and adolescent well-being are small on average and highly context-dependent, with risk concentrated among heavy or problematic use and among those already vulnerable [91]. Likewise, a meta-analysis of longitudinal studies indicates that loneliness and problematic media use can reinforce each other over time, supporting potential bidirectional pathways [92].

The evidence base is limited by the predominance of cross-sectional designs, reliance on self-reported exposure, inconsistent definitions of 'use' (time, platform, content, and interaction quality), and residual confounding. Future studies should incorporate objective digital trace measures, examine key moderators (e.g., active versus passive use, content type), and evaluate longer-term outcomes and interventions that promote healthy digital social participation.

4 Discussion

This narrative review synthesizes evidence on how SMAT influences the six pillars of lifestyle medicine. Across all pillars, a consistent pattern emerges: digital technologies offer significant potential benefits for health behavior change and disease prevention, but also introduce novel risks and challenges that must be carefully managed (Table 1).

4.1 Evidence quality and limitations

A critical appraisal of the studies included reveals several important limitations that must be considered when interpreting findings and making clinical recommendations.

- Sample size and statistical power

Significant percent of the included randomized controlled trials had sample sizes below 100 participants, limiting statistical power to detect meaningful effects and generalizability of findings. Pilot studies and feasibility trials were common, particularly for novel

Table 1 Summarizes the main positive and negative impacts of SMAT for behaviour change,

Pillar	Positive impacts of SMAT	Negative impacts of SMAT
Nutrition	Access to nutrition education; self-monitoring of intake; personalised feedback; gamified apps that increase fruit/vegetable intake; online communities that support healthy eating; personalised digital nutrition interventions (e.g., Food4Me)	Exposure to fad diets and misinformation; body image pressures; disordered eating; emotional eating; promotion of ultra-processed food via delivery apps and digital marketing; strong impact on children's food preferences
Physical activity	Wearables and apps that track steps and activity; smartphone interventions that increase MVPA and reduce sedentary time; active video games for youth; social media challenges and peer support that foster adherence; gamification (badges, points, leaderboards)	Overreliance on devices; obsession with numbers (steps, calories); anxiety when goals are not met; unrealistic fitness portrayals leading to discouragement; displacement of outdoor or unstructured play
Stress management	Digital mental health programmes, therapy apps, and telehealth; mindfulness and relaxation apps; online peer-support communities; improved cognitive stimulation and engagement for older adults; small but significant reductions in distress in some interventions	Technostress, digital fatigue, and burnout; cyberbullying and harassment; social comparison and low self-esteem; overuse leading to sleep disruption; reliance on unregulated apps of variable quality
Substance use	Social media and text-based smoking cessation campaigns; targeted digital public health messaging; Facebook and mobile interventions that improve quit rates; online peer-support groups for abstinence	Glamorisation and normalisation of tobacco, alcohol, and cannabis; high proportion of substance-related content portraying use positively; influencer and industry marketing targeting youth; dose–response relationship between screen time and experimentation or use
Restorative sleep	Wearables and apps that track sleep duration and quality; digital sleep hygiene education; blue-light filters and bedtime modes; behavioural sleep interventions delivered via apps	Bedtime screen use delaying sleep onset; reduced deep sleep; melatonin suppression; nighttime notifications and FOMO; poorer sleep quality and satisfaction in adolescents and adults; associations with depression and anxiety
Social connection	Maintenance of relationships across distance; support for marginalised groups and rare conditions; online communities that reduce loneliness; opportunities for creative expression and identity exploration; especially useful during crises (e.g., COVID-19)	Time displacement from family, school, or offline friendships; superficial interactions replacing deeper ties; increased loneliness and perceived social isolation at high levels of use; social comparison, envy, and reduced self-esteem; potential erosion of face-to-face communication, especially in older adults

digital interventions, reflecting the rapid pace of technological innovation outpacing rigorous evaluation.

- Short follow-up periods

The median follow-up duration across intervention studies was 8–12 weeks. This limitation is particularly concerning for lifestyle medicine, where sustained behavior change over years is the ultimate goal. Short-term efficacy does not guarantee long-term effectiveness, and the "novelty effect" of new technologies may inflate short-term outcomes. Research on sustained engagement and long-term health outcomes is urgently needed.

- Reliance on self-report measures

Most of the studies relied primarily or exclusively on self-reported outcomes (dietary intake, physical activity, sleep quality, stress levels, social connection), which are subject to recall bias, social desirability bias, and measurement error. Self-report may inflate effect sizes, particularly for socially desirable behaviors. While objective measures (accelerometry, biomarkers, polysomnography, ecological momentary assessment) were used in some studies, they remain the minority. Future research should prioritize objective outcome assessment whenever feasible.

- Selection and participation bias

Many digital health studies recruited participants who were already motivated, tech-savvy, health-conscious, or actively seeking behavior change interventions. This selection bias limits generalizability to broader populations, including those with limited digital literacy, low motivation, or significant barriers to technology access. Limited representation of older adults, low-income populations, racial and ethnic minorities, and individuals with disabilities in digital health research is a significant concern for health equity. Differential attrition that exceeds 20% (e.g., less engaged participants dropping out) may bias results toward overestimating effectiveness.

- Heterogeneity in interventions and outcomes

Wide variation exists in the types of digital interventions studied (smartphone apps, wearable devices, social media platforms, web-based programs, text messaging, telemedicine), intervention components (education, self-monitoring, feedback, social support, gamification), and implementation approaches (standalone vs. integrated with human support). This heterogeneity limits comparability across studies and makes synthesis challenging.

Inconsistent outcome measures across studies further complicate evidence synthesis. For example, physical activity has been measured as steps per day, minutes of moderate-to-vigorous activity, metabolic equivalents, or self-reported exercise frequency, each capturing different aspects of activity behavior. Lack of standardized assessment tools for digital health literacy, user engagement, and intervention fidelity also limits cross-study comparisons.

- Publication bias

Potential overrepresentation of positive findings exists, as studies with null or negative results are less likely to be published. Industry-sponsored research may be subject

to conflicts of interest, with potential bias toward favorable findings for commercial products.

4.2 Synthesis across pillars and evidence gaps

Across pillars, common mechanisms include self-monitoring, just-in-time prompts, feedback loops, gamification, and social reinforcement. When these features are grounded in behaviour change theory and integrated with coaching, they can improve engagement and short-term outcomes (e.g., steps and dietary tracking) [34–36]. However, the same design features can encourage compulsive checking and digital fatigue, while algorithmic curation can amplify misinformation, commercial bias, and appearance-focused norms that undermine health literacy and well-being.

- Data privacy and safety are central concerns. Many commercial platforms collect granular behavioral and biometric data, and opaque data-sharing practices may expose users to targeted advertising, profiling, or discriminatory uses.
- Vulnerable populations,—such as children and adolescents, people with mental health conditions, and individuals living with chronic diseases— may be at increased risk of dangers of data misuse, cyberbullying, online harassment, or exposure to harmful content (e.g., those being pro-eating disorders/ pro-substance use communities). In young people, an appealing user interface and algorithmic recommendations can drive up compulsive use of a platform, distort perceptions of risk, and erode autonomy. Health care Providers, policymakers and advocates should push for age-appropriate design, robust content moderation and protection from exploitative marketing. (Table: 2)
- Equity is another key consideration. The 'digital divide' means that older adults, people with disabilities, and those with lower socioeconomic status or living in underserved areas may have less access to reliable devices, connectivity, and digital literacy support. Digital interventions should therefore be designed for accessibility, cultural relevance, and low-bandwidth settings, and should be offered alongside non-digital alternatives.

Finally, the current evidence base is constrained by heterogeneous interventions, small samples, short follow-up, and frequent reliance on self-reported exposure and outcomes. Observational studies also face confounding (e.g., baseline mental health and social context), and few trials evaluate sustained clinical outcomes. Future work should prioritize prospective designs, objective measures (including device and platform data where appropriate), standardized outcomes across pillars, and robust evaluation of long-term effectiveness, harms, and cost-effectiveness in real-world settings.

4.3 Action-oriented recommendations

The following recommendations provide actionable, evidence-based advice on how to use digital tools to support healthy living for adults.

1. Intentional and purposeful technology use

   Practice digital mindfulness

   - Limit usage to one device or app at a time to help avoid digital overload.
   - Keep recreational screen time (aside from work) to ≤ 2 h/day.

- Avoid working while your favorite movie or show is playing in the background [93].
- Reflect on personal usage habits weekly to increase awareness and adjust accordingly.
- Disconnect before sleep
- Eliminate screen exposure 1–2 h prior to bedtime to preserve melatonin levels and sleep quality [94].
- Keep phones out of the bedroom and turn on blue light filters if you must use them after hours.
- Promote a digital timeout
- Use the 20-20-20 rule: every 20 min, focus at least 20 feet away for at least 20 s to ease eye strain.
- Add 5–10-minute breaks at the end of the hour and assign yourself at least one screen-free day or evening each week to allow for mental restoration and social connection [95].

2. Wellness interventions powered by advanced technologies

Use health-enhancing apps and devices

- Interact with fitness trackers and health Apps to track steps taken, time spent sleeping, fluid intake, heart rate, or mental health [96–98].
- Set personalized goals using features available to help with motivation and compliance.
- Purposeful engagement with digital platforms
- Promote content that improves physical, emotional, or social well-being.
- Take part in exercise and nutrition tracking, education, and mindfulness sessions via platforms [99, 100].
- Set boundaries between work and home
- Set working time and turn off notifications outside work [101].
- Choose alerts to receive only the most relevant information and minimize distractions [102, 103].

3. Improving lifestyle via social media

Prioritize media literacy

- Follow content from trusted public health organizations, clinicians, or educators [104–106].
- Approach health advice found online with skepticism, particularly if there are no citations or disclosure of where the advice comes from.
- Supportive online community participation
- Participate in digital communities around common health objectives (e.g., plant-based diets, running challenges, or mental wellness forums [107–109].
- Having peers for support can promote accountability, motivation, and emotional resilience.
- Health goal setting on social media
- Share goals publicly, like daily steps taken or healthy meals eaten, and seek support and feedback.

Table 2 Essential "Ms" for media use in school-aged children and adolescents

Category (The 4 Ms)	Recommendations
Manage Screen Time	Create and update a Family Media Plan with clear limits on screen time and content - Engage with children during screen use (co-view or co-play) - Discourage multitasking during homework - Use parental controls and privacy settings - Have access to children's device and social media login info - Discuss online behaviour and safety regularly
Meaningful Media Engagement	Prioritize sleep, physical activity, and face-to-face interaction over screen use - Encourage educational, social, or active screen activities - Help children choose age-appropriate content - Participate in children's media activities (e.g., gaming) - Promote digital literacy programs in schools and child care settings
Model Positive Media Habits	Encourage parents to assess and adjust their own media habits - Promote offline hobbies and outdoor play - Warn about risks of using devices while walking, biking, or driving - Designate screen-free times (e.g., during meals) - Keep screens off when not in use - Avoid screens 1 h before bed and limit use in bedrooms
Monitor for Harmful Use	Look for signs like boredom or distress without devices - Watch for defiance of screen-time limits - Identify if screen use disrupts sleep, school, or relationships - Observe emotional distress linked to online or gaming experiences

- Engage in hashtag campaigns to build more engagement and habits [110, 111].
- Critic health content and know your purpose
- Validate claims through reputable sources to avoid misinformation.
- Explore alternative formats such as infographics and short-form videos [112, 113].

4. Encouraging balance and mental health

Monitor the use of social media

- High social media usage is associated with depression and anxiety [114, 115].
- Promote balance by trading some digital time for physical activity, sleep, or another form of meaningful in-person interaction [116, 117].

Advocate for enduring digital ecosystems

- Follow organizations that advocate for health and transparency.
- Report wrongly and block toxic content [104, 118].
- Every Family should be counselled to remember four essential 'Ms' for school age and adolescents:

Every Family should be counselled to remember four essential 'Ms' for school age and adolescents: (Table 2)

- Ponti [119].

5 Conclusion

SMAT can augment lifestyle medicine by enabling scalable self-monitoring, education, and support across the six pillars; however, effects are not universal and depend on the quality of content, platform design, and patient context.

For health care providers, practical steps include: (1) clarifying patient goals and digital readiness; (2) recommending evidence-based tools and integrating them with behaviour counselling and follow-up; (3) screening for harms such as sleep disruption, problematic use, anxiety, and exposure to misinformation; and (4) promoting boundaries that

protect sleep, movement, and in-person relationships. For educators and policymakers, priorities include strengthening digital health literacy, protecting children and other vulnerable groups, improving transparency and data governance, and investing in rigorous evaluation and equitable access to effective digital health tools.

In conclusion, SMAT have great potential as tools of behaviour change, social support, and health promotion, provided they are used mindfully. Setting digital boundaries, enhancing literacy, and integrating tech-native self-monitoring into lifestyle can go a long way to make life more meaningful. The aim, then, is to transition from passive to active, intentional use that contributes to well-being over time.

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L.A. wrote the main manuscript. share in the conception process.A.A.K.: Idea conception, supervision during writing phases, and revision of the entire document.R.A.: Revision; preparing a table. She actively participated in the conception process.H.A.: Logistic support. She thoroughly reviewed the entire document and carried out the final editing.All Authors: shared in responding to reviewers' comments.

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No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This is a narrative review article that synthesizes and scrutinizes previously published literature. No primary data collection was conducted as part of this study. Therefore, ethical approval from an institutional review board was not required. All studies cited in this review were published in peer-reviewed journals and had obtained appropriate citations as reported in their original publications. This narrative review did not involve any primary research with human participants. All information included in this manuscript were obtained from publicly available literature.

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